COURSE OUTLINE

1. Course Title: System Analysis and Design

2. Course Code: BIIT 1303

3. Credit Value: 3

4. Course Synopsis: This course introduces students to the principles and practices of system analysis and design, emphasising the concepts and methods used for developing effective information systems, with a focus on the object-oriented approach. Students will learn how to improve the operations and workflows of an organisation through the development of information systems. Emphasis will be placed on analysing business requirements, modelling system components, and designing solutions that meet organisational needs, aiming to improve business processes.

5. SAF Elements: This course is designed to enhance students' problem-solving and analytical skills, encouraging them to develop innovative solutions. Through the course activities and assessments, students not only acquire knowledge but also demonstrate responsibility (Amanah) by applying their knowledge to develop information system designs. The emphasis on learning (Iqra') prompts students to seek knowledge actively and apply it innovatively, fostering a culture of continuous learning and ethical responsibility.

6. Course Classification within the Curriculum: Major (core)

7. Pre-requisite(s) (if any): BICS 1301 Elements of Programming

8. Co-requisite(s) (if any):

9. Course Learning Outcomes:

No.	Learning Outcomes (CLO)	Programme Outcomes (PLO)
1	Describe major alternative methodologies in information system development, business issues and feasibility analysis (C2, PLO2)	PLO2
2	Produce a system prototype by applying requirements gathering techniques, and preparing requirement definitions and system design specifications. (P4, PLO3)	PLO3

No.	Learning Outcomes (CLO)	Programme Outcomes (PLO)
3	Demonstrate effective communication skills by articulating technical concepts and analytical insights, collaborating with team members to propose a system prototype. (A3, PLO5)	PLO5

10. Constructive Alignment:

CLO	Teaching-Learning Methods	Assessment Methods
CLO1	Lecture/Tutorial	Continuous Assessment, Final Examination
CLO2	Lecture/Tutorial	Continuous Assessment
CLO3	Project - based learning	Continuous Assessment

11. Assessment Distribution:

Assessment Methods	Percentage
Must-pass Assessment Method(s)	**Percentage**
Continuous Assessment	60
Final Examination	40
Total	**100**

12. Course Content

Week	Course Content	Guided Learning SLT	Independent Learning SLT
1	Systems, Roles, and Development Methodologies -Types of Systems -Need for Systems Analysis and Design -Roles of the Systems Analyst -The Systems Development Life Cycle	3	3
2	Understanding and Modeling Organizational Systems -Organizations as Systems -Levels of Management -Organizational Culture	3	4
3	Project Management -Project Initiation -Determining Feasibility -Ascertaining Hardware and Software Needs -Managing the Project	3	4
4	Project Management -Group Project discussion	3	4
5	Information Gathering: Interactive Methods -Interviewing -Using Questionnaires	3	4

Week	Course Content	Guided Learning SLT	Independent Learning SLT
6	Information Gathering: Unobtrusive Methods -Sampling -Observing a Decision Maker's Behavior -Observing the Physical Environment	3	4
7	Agile Modeling and Prototyping -Prototyping -Developing a Prototype -Agile Modeling	3	4
8	Object-Oriented Systems Analysis and Design Using UML -Object-Oriented Concepts -Use Case Modeling	3	5
9	Object-Oriented Systems Analysis and Design Using UML -Activity Diagrams -Class Diagrams	3	5
10	Designing Effective Output -Output Design Objectives -Relating Output Content to Output Method	3	5
11	Designing Effective Input -Good Form Design -Good Display and Web Forms Design	3	4

Week	Course Content	Guided Learning SLT	Independent Learning SLT
12	Human-Computer Interaction -Understanding Human-Computer Interaction -Usability -Types of User Interface -Special Design Considerations for Ecommerce	3	4
13	Group Project Discussion	3	8
14	Project presentation	3	8
	Final Assessment (if applicable)	3	9
	Total	45	75

13.0. References:

13.1 Required:

1) Kendall, K. E., & Kendall, J. E. (2023). Systems Analysis and Design.

Prepared by:	**Checked by:**
Name: Asst. Prof. Dr. Elin Eliana Abdul Rahim Department/Unit: Information Systems Date: 4 June 2024	Name: Asst. Prof. Dr. Mohd Khairul Azmi B Hassan Head: Department of Information Systems Date: 4 June 2024

ANNEX

I. Course Plan

Notes:

1. ***The course plan is to be prepared and reviewed semesterly before new semester starts.***
2. ***Changes to the course plan do not require submission to the Senate***

*** Can be according to class session, or week depending on the nature of the course*

$ *May include* any information to assist the learning activities

<table>
<tr><td>COURSE TITLE</td><td colspan="4">System Analysis and Design</td><td>COURSE CODE</td><td>BIIT 1303</td></tr>
<tr><td>CREDIT HOUR</td><td colspan="4">3</td><td>SLT</td><td>120</td></tr>
<tr><td>INSTRUCTOR</td><td colspan="6">Asst. Prof. Dr. Elin Eliana Abdul Rahim
Assoc. Prof. Ts. Dr. Zahidah Zulkifli</td></tr>
<tr><td colspan="2" rowspan="2">CONTENT AND MAIN ACTIVITIES (Topics, Teaching-Learning Activities)</td><td colspan="3">SLT DISTRIBUTION</td><td colspan="2" rowspan="2">$REMARKS</td></tr>
<tr><td>Synchronous</td><td>Asynchronous</td><td>Independent Learning</td></tr>
<tr><td>1</td><td>Systems, Roles, and Development Methodologies
-Types of Systems
-Need for Systems Analysis and Design
-Roles of the Systems Analyst
-The Systems Development Life Cycle</td><td>3</td><td></td><td>2</td><td colspan="2"></td></tr>
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2	Understanding and Modeling Organizational Systems -Organizations as Systems -Levels of Management -Organizational Culture	3		3	
3	Project Management -Project Initiation -Determining Feasibility -Ascertaining Hardware and Software Needs -Managing the Project	3		3	
4	Project Management -Group Project discussion	3		3	
5	Information Gathering: Interactive Methods -Interviewing -Using Questionnaires	3		3	
6	Information Gathering: Unobtrusive Methods -Sampling -Observing a Decision Maker’s Behavior -Observing the Physical Environment	3		3	
7	Agile Modeling and Prototyping -Prototyping -Developing a Prototype -Agile Modeling	3		3	

8	Object-Oriented Systems Analysis and Design Using UML -Object-Oriented Concepts -Use Case Modeling	3		4	
9	Object-Oriented Systems Analysis and Design Using UML -Activity Diagrams -Class Diagrams	3		4	
10	Designing Effective Output -Output Design Objectives -Relating Output Content to Output Method	3		4	
11	Designing Effective Input -Good Form Design -Good Display and Web Forms Design	3		4	
12	Human-Computer Interaction -Understanding Human-Computer Interaction -Usability -Types of User Interface -Special Design Considerations for Ecommerce	3		4	

13	Group Project Discussion	3		4	
14	Project presentation	2		4	
	Continuous Assessment				
1	Quiz			4	
2	Presentation	1		4	
3	Project			10	
	Final				
1	Final Examination	3		9	
TOTAL SLT					120